

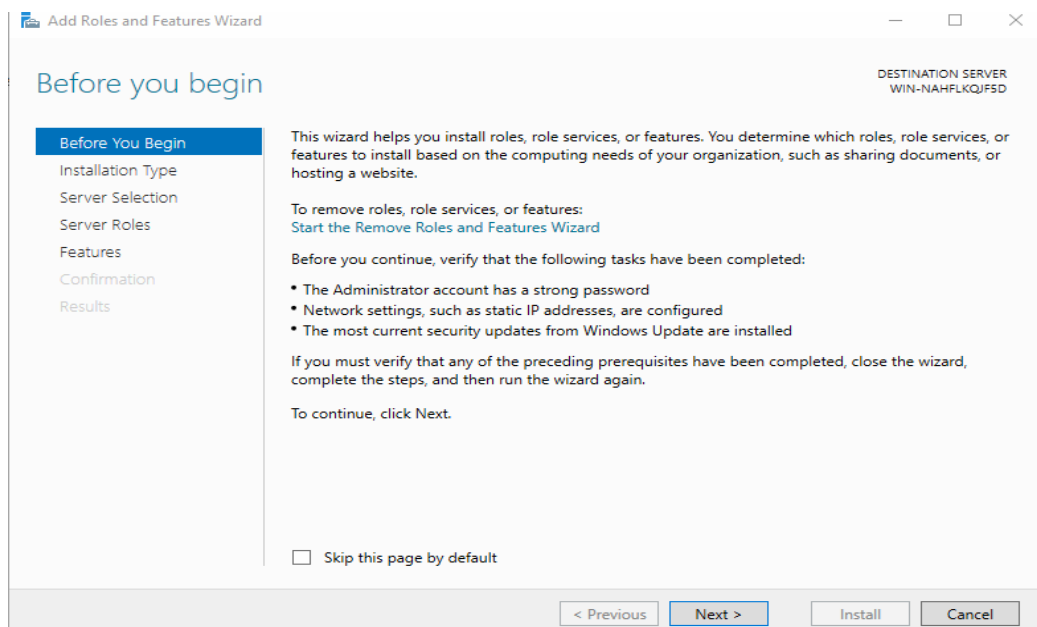
7a) Configuration of HyperV on Windows of VMs

Aim:

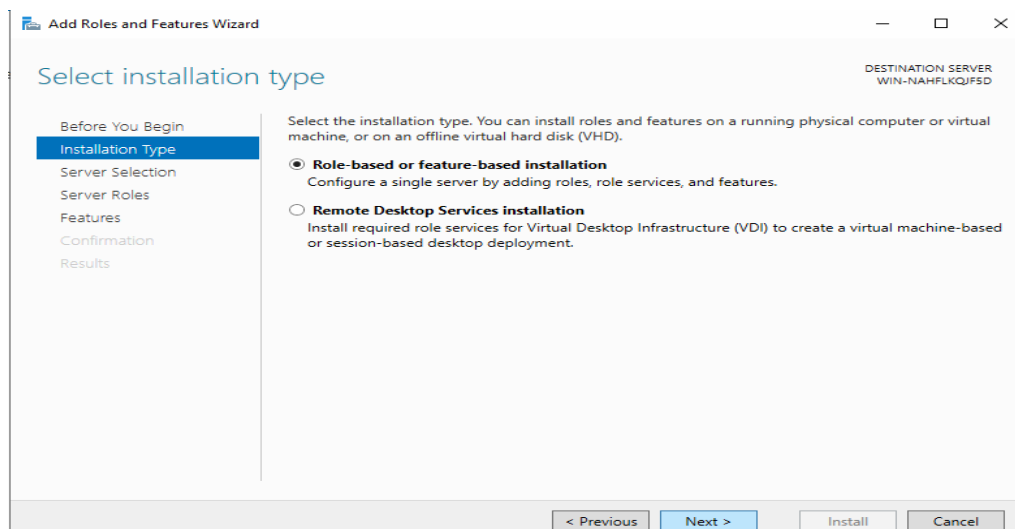
To configure Hyper-V on Windows for creating and managing virtual machines (VMs), enabling efficient virtualization and resource utilization.

Steps:

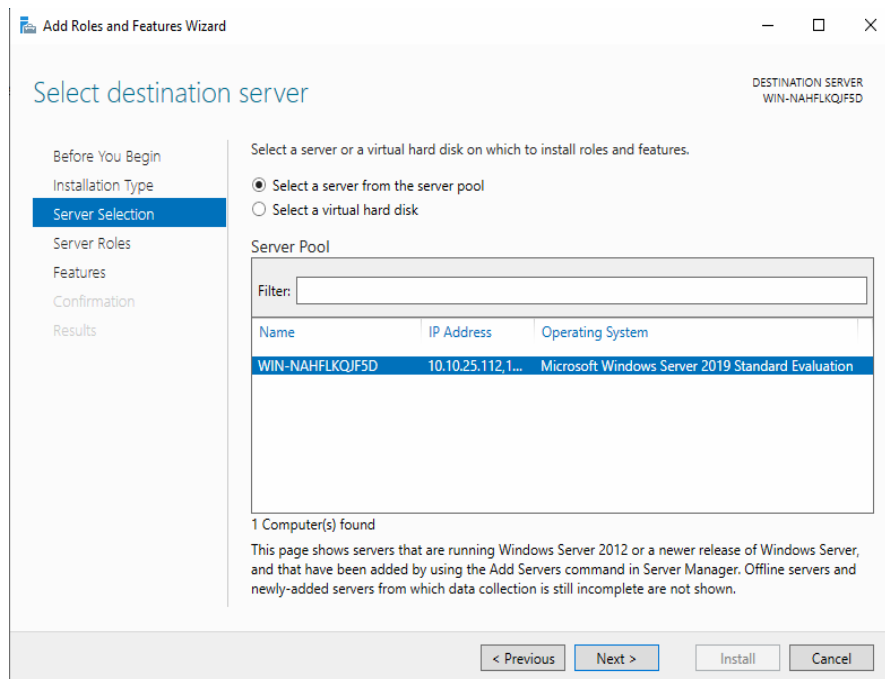
1) Open the windows server and open the server manager Click add roles and features . Read before you begin and click next.



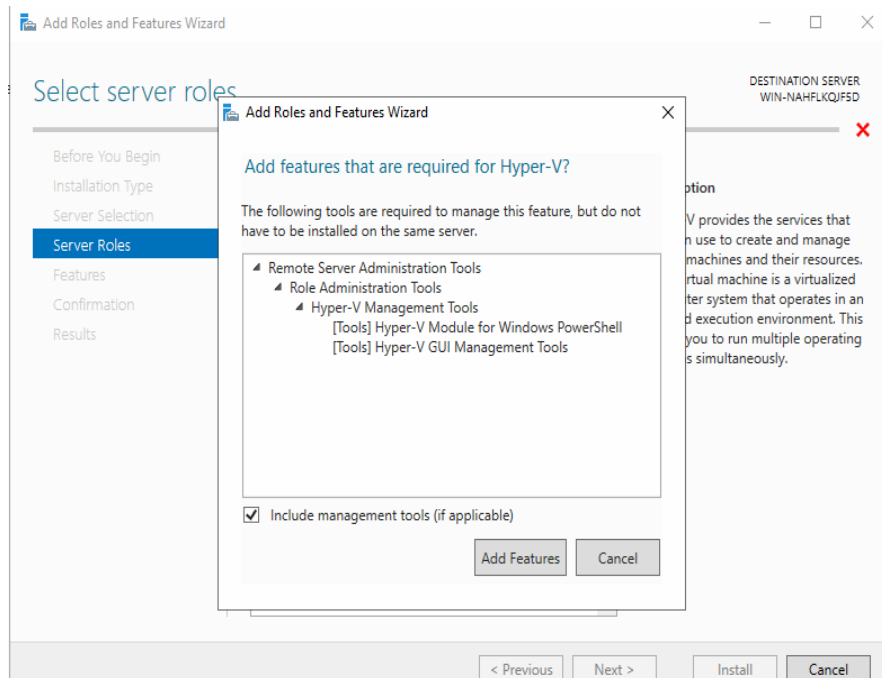
2) Select role based installation type and click next.



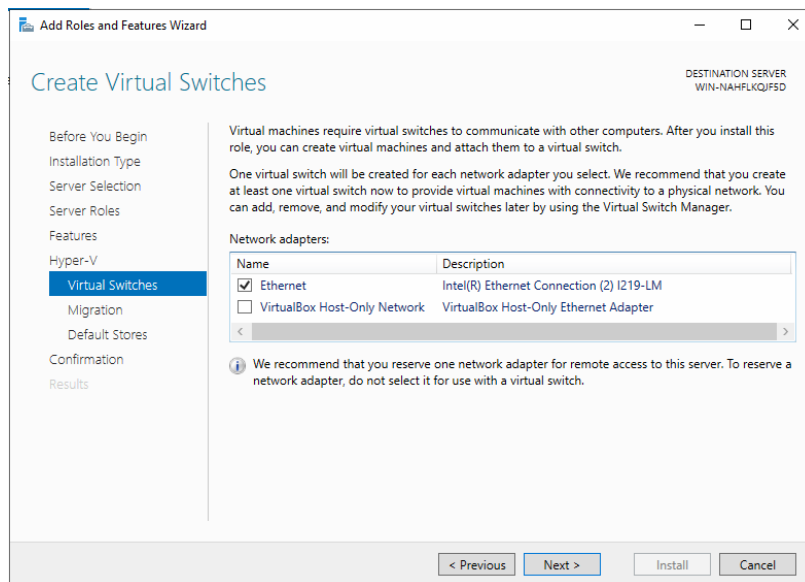
3) Select the server and click next



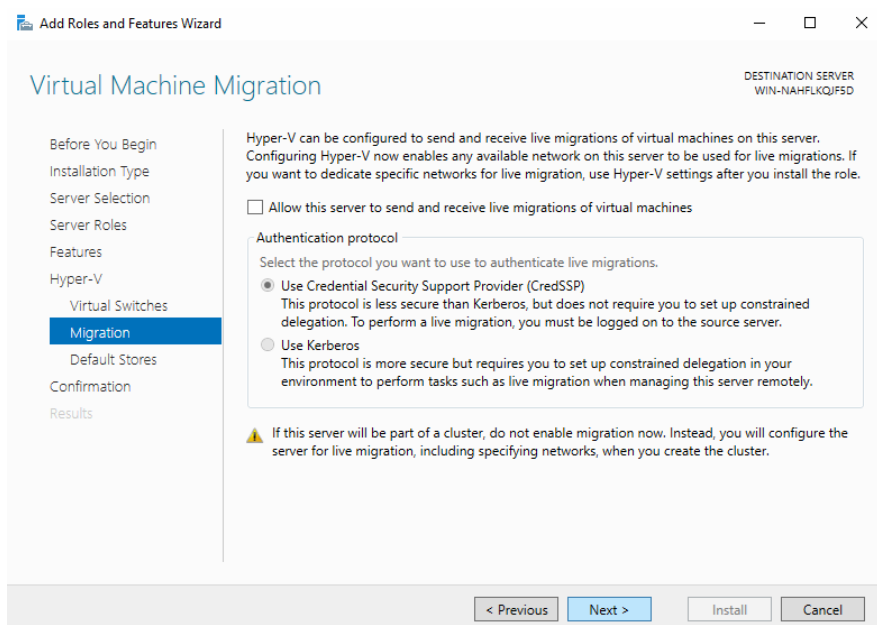
4) In server roles .check on hyper v and click the add features .



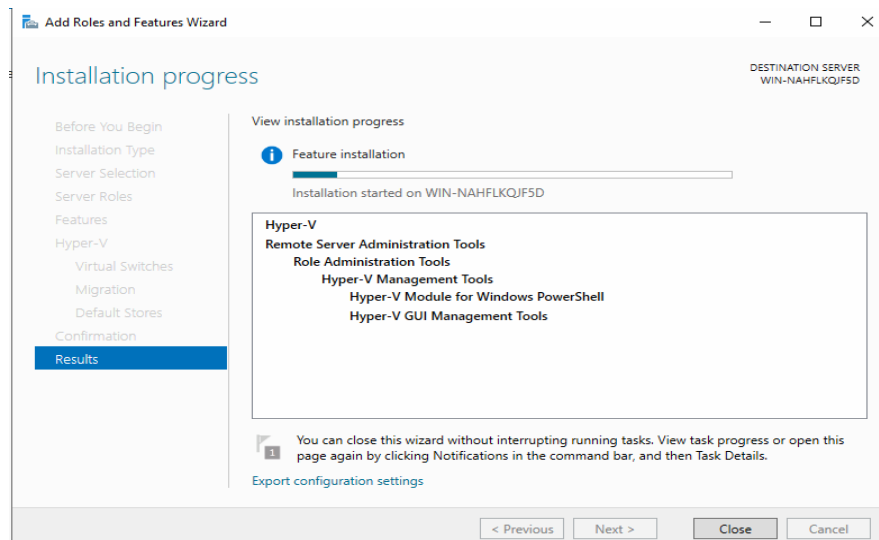
5) Enable virtual switches both ethernet and virtual box host only adapter



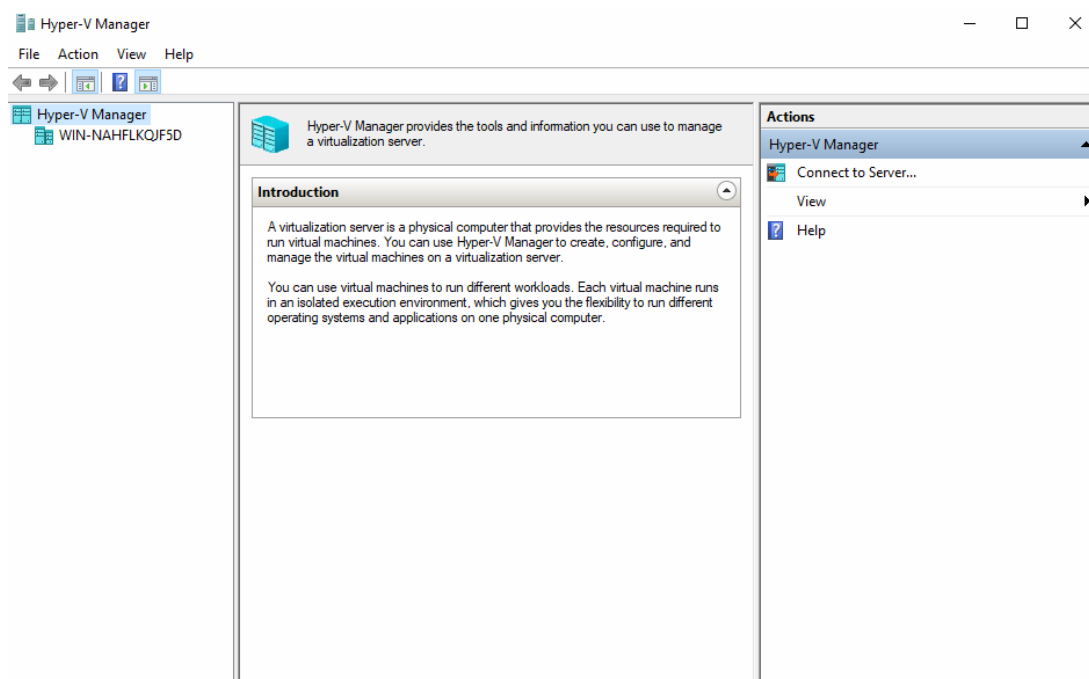
6) For migration use credential security support provider



7) Click install and hyperv has been installed and enabled on the windows server



8) Open hyperv manager in tools



RESULT:

Thus the configuration Hyper-V on Windows for creating and managing virtual machines (VMs), enabling efficient virtualization and resource utilization has been executed successfully.

7b) Creation of VM, Powershell commands, Management

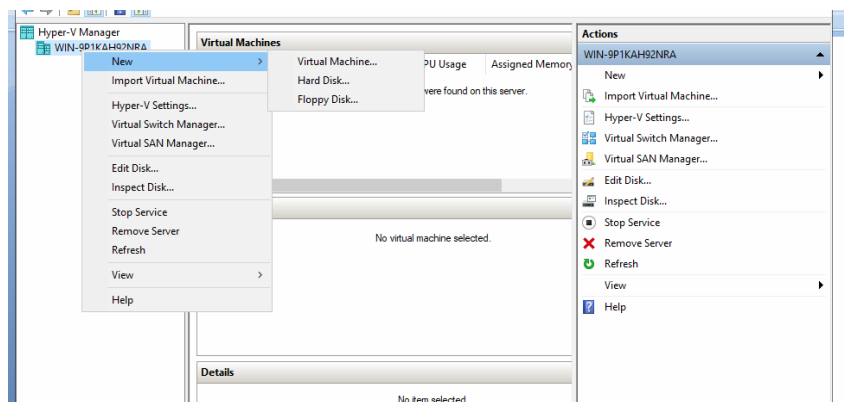
Aim:

To create and manage Virtual Machines (VMs) using Hyper-V, including PowerShell commands for automation and efficient administration.

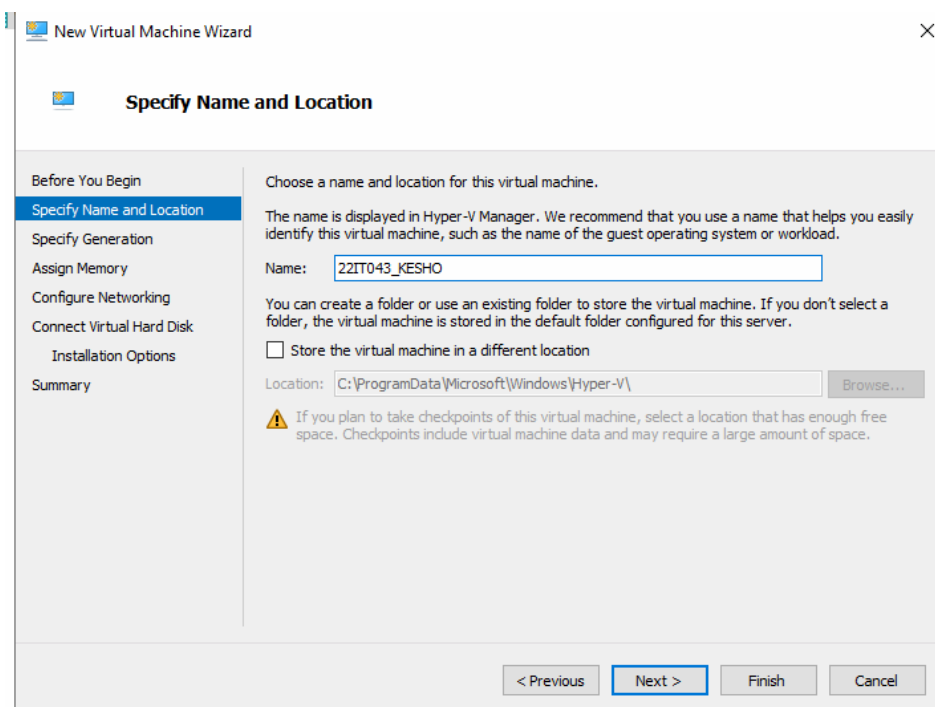
Procedure:

(Vm installation without powershell)

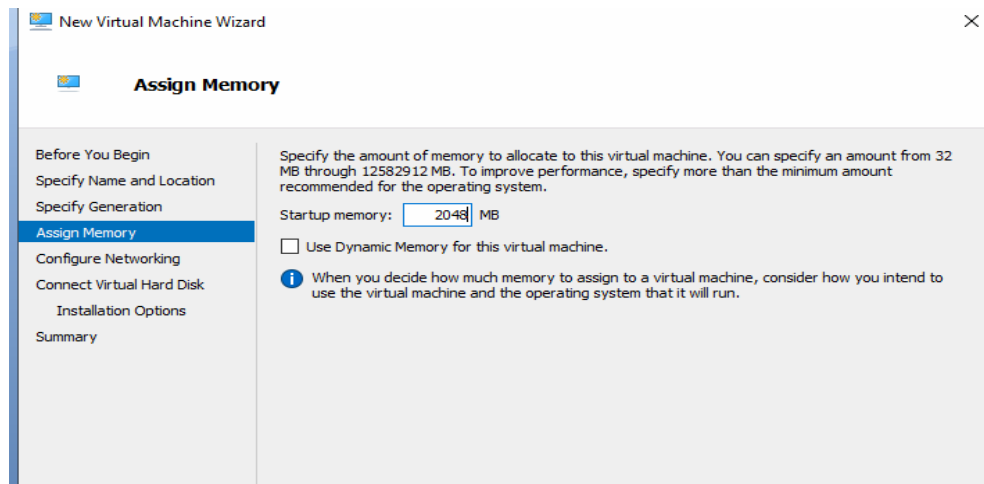
1)open Tools-> hyper v manager-> right click on server and click create new vm



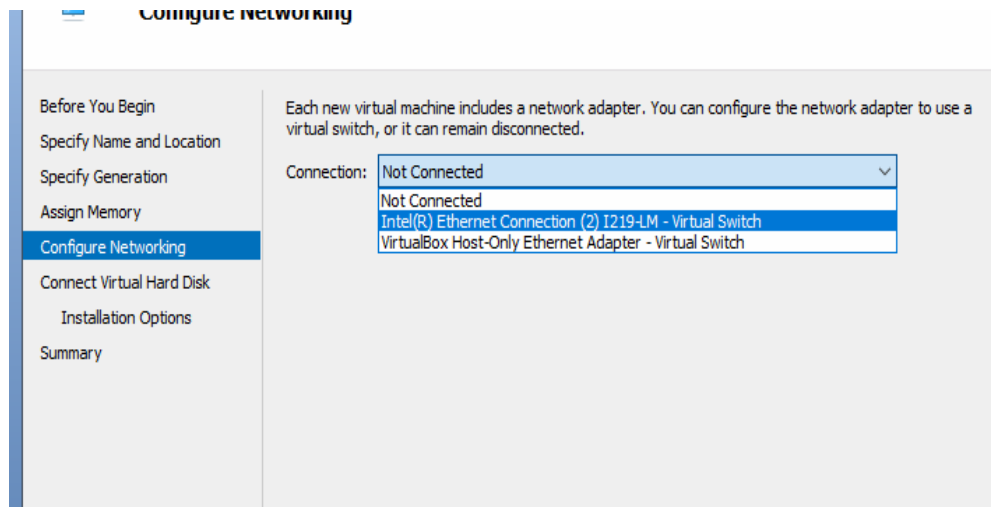
2)Specify name and location click next.



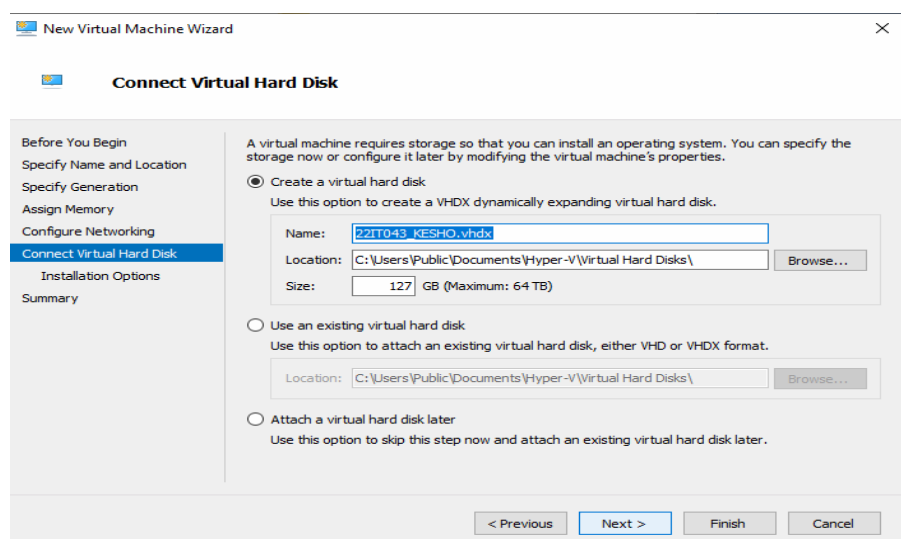
3)Assign the memory for the vm(for safety give as 2gb)



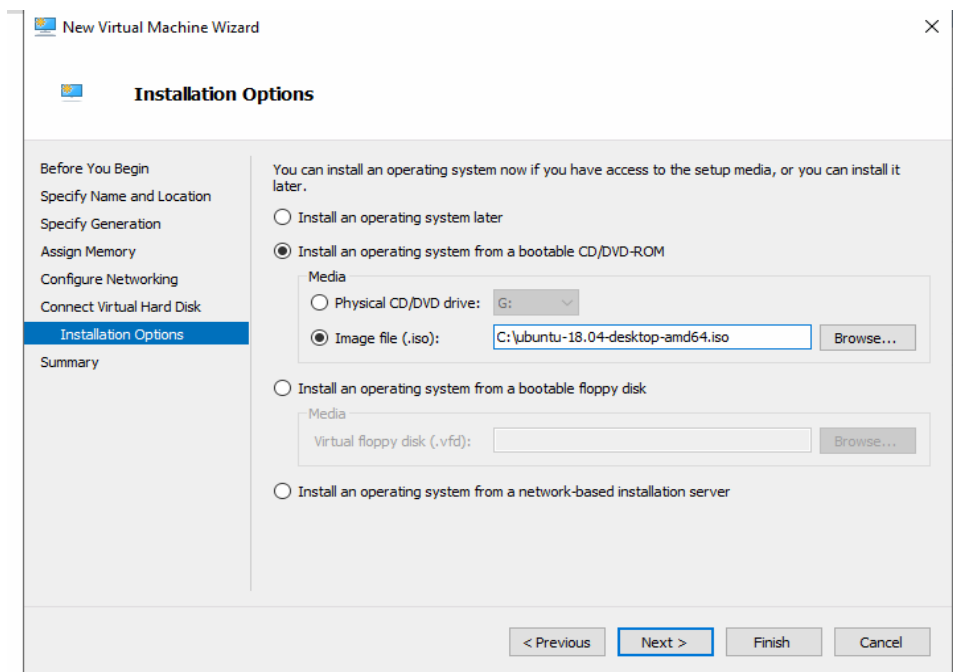
4)Configure the network click on interR Ethernet connection virtual switch



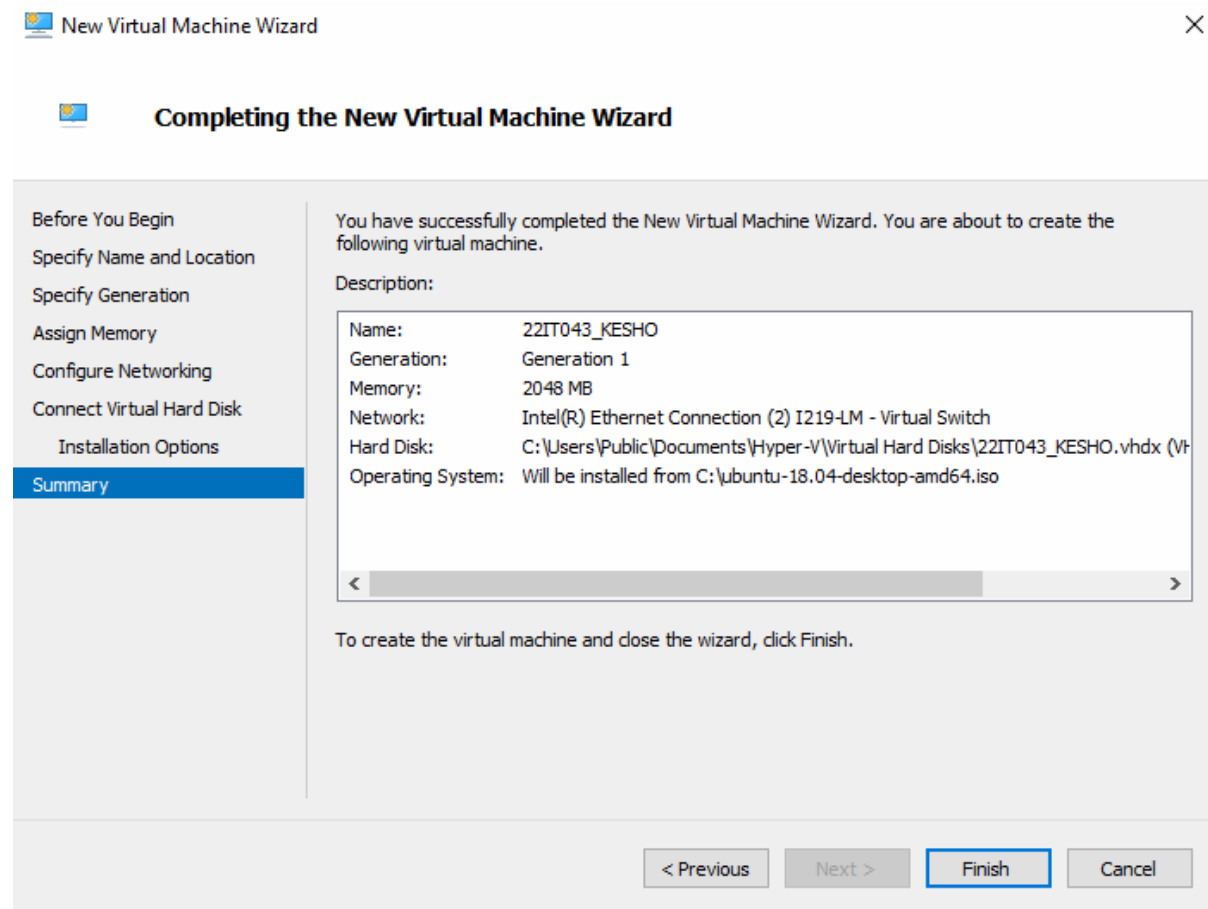
5)Connect virtual harddisk (name,location,maximum gb) click next



6) Import the iso file which os need to install on the vm(here ubuntu)



7) Click finish



Vm installation using Powershell **Procedure :**

- 1) Create vm with name and space in gb with virtual hard disk creation

```
PS C:\Users\Administrator> New-VM -Name "tcevm" -MemoryStartupBytes 2GB -NewVHDPATH "D:\tcevm.vhdx" -NewVHDSIZEBytes 60GB -Path "D:\\"
Name State CPUUsage(%) MemoryAssigned(M) Uptime Status Version
-----
tcevm Off 0 0 00:00:00 Operating normally 9.0

PS C:\Users\Administrator> Get-VM
Name State CPUUsage(%) MemoryAssigned(M) Uptime Status Version
-----
22IT043_KESHO Off 0 0 00:00:00 Operating normally 9.0
tcevm Off 0 0 00:00:00 Operating normally 9.0

PS C:\Users\Administrator>
```

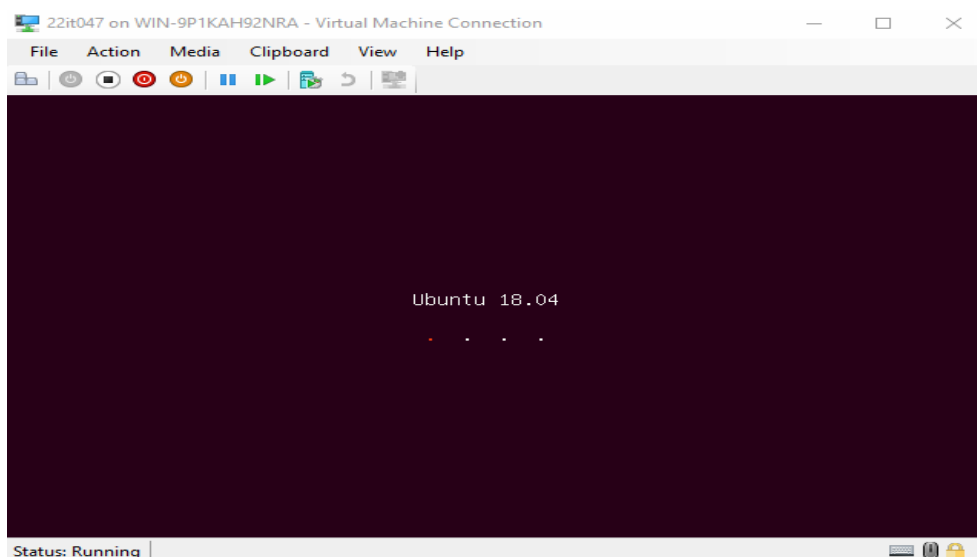
- 2) Assign virtual switch to the vm

```
PS C:\Users\Administrator> Connect-VMNetworkAdapter -VMName "tcevm" -SwitchName "Intel(R) Ethernet Connection (2) I219-L
M - Virtual Switch"
PS C:\Users\Administrator>
```

- 3) import the iso file which os need to install on vm

```
PS C:\Users\Administrator> Connect-VMNetworkAdapter -VMName "tcevm" -SwitchName "Intel(R) Ethernet Connection (2) I219-L
M - Virtual Switch"
PS C:\Users\Administrator> Set-VMdvdDrive -VMName "tcevm" -Path "C:\ubuntu-18.04-desktop-amd64.iso"
PS C:\Users\Administrator>
```

- 4) Start the vm



Powershell commands

1)Get vm and start vm

```
PS C:\Users\Administrator> Get-VM

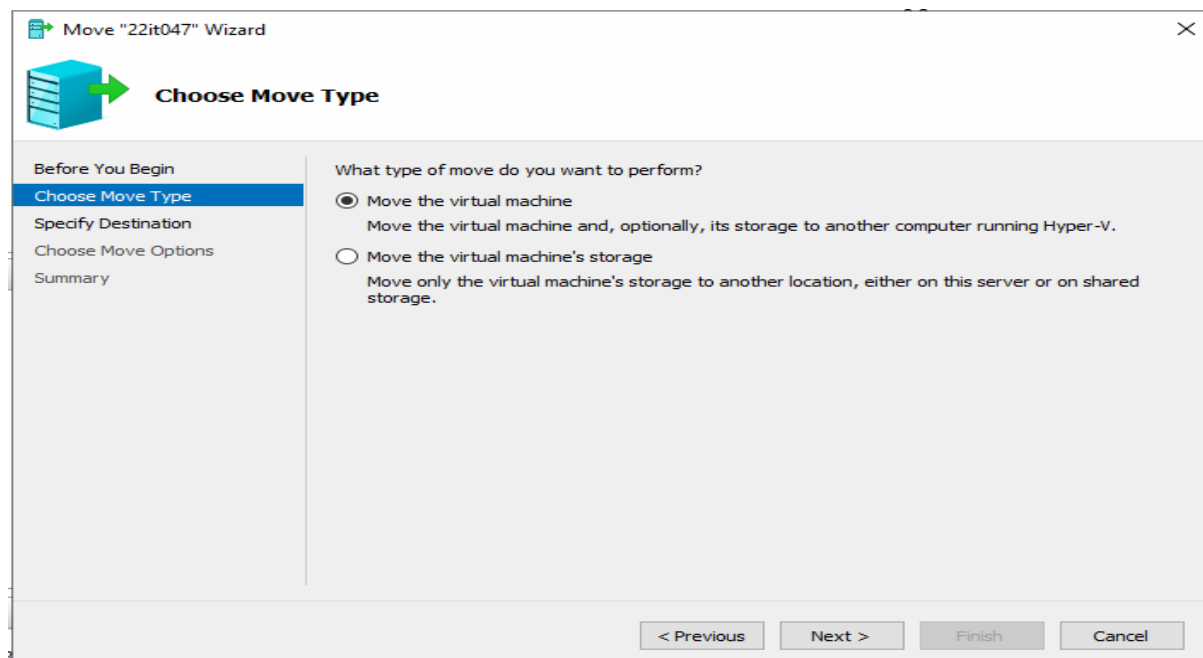
Name                State CPUUsage(%) MemoryAssigned(M) Uptime        Status           Version
-----
22IT043_KESHO      Off    0           0                00:00:00 Operating normally 9.0
22it047            Off    0           0                00:00:00 Operating normally 9.0
tcevm              Off    0           0                00:00:00 Operating normally 9.0

PS C:\Users\Administrator> Start-VM -Name "22it047"
```

2)Stop vm

```
PS C:\Users\Administrator> Stop-VM -Name "tcevm"
PS C:\Users\Administrator>
```

Migration(Right click the vm and click move) and give finish .



Result:

Thus the creation and manage Virtual Machines (VMs) using Hyper-V, including PowerShell commands for automation and efficient administration has been executed successfully.